QUANTUM MACHINE LEARNING · RISING STAR

Entangled Futures

Quantum Kernel Outage Escalation Study

TEAM	Entangled Futures	PARTICIPANT	Kamilah Sanders
AFFILIATION	Greater Than Equal	DEPARTMENT	Not applicable — independent industry participant
CAREER LEVEL	Industry professional, CEO / Global Ecosystem Strategist	SUBMISSION CATEGORY	Rising Star
TRACK	Quantum Machine Learning	DATE	October 24, 2026

RESEARCH QUESTION

During an ongoing reported outage, can recent outage counts and previously issued storm warnings help rank county-hours by the likelihood of a substantial increase in reported customer outages 60–75 minutes later?

Why this matters

When electricity outages worsen, communities may need additional support to keep people safe, fed, and connected. Emergency coordinators must decide where to focus attention and prepare resources, often with incomplete information.

This project investigates whether outage histories and previously issued storm warnings can help identify places where reported outages may escalate. Such information could prompt coordinators to investigate local needs and consider additional support.

Outage counts indicate electricity disruption; they do not identify who is suffering or how well a community is supporting its residents. Any future operational use would require local knowledge, community relationships, and information about actual needs.

This initial feasibility study evaluates whether outage histories and previously issued weather warnings could provide useful information for emergency coordination. It examines prediction performance and limitations; it does not evaluate coordination decisions or outcomes for people.

Summary of work

- Audited South Carolina's 2022 ORNL EAGLE-I records for reporting continuity, duplicate conflicts, explicit zeros, and candidate escalations.
- Defined an outcome and constructed four historical outage-count inputs without using future counts as predictors.
- Retrieved original National Weather Service severe-thunderstorm and tornado warning messages from the Iowa Environmental Mesonet archive and extracted the counties named at issuance.
- Compared simple rules, logistic regression, random forests, a conventional SVM, and a Qiskit quantum-kernel SVM.
- Used chronological partitions and a matched 639-example training sample for the two SVMs.
- Reproduced the final results in a separate cleaned Colab notebook.

Motivation for the QML track

My work connects energy systems, emerging technologies, and community resilience. I chose quantum machine learning to gain practical experience with the technology and investigate its potential relevance to the challenges utilities and communities face.

I am particularly interested in how information can support coordination when electricity disruptions threaten people's safety and access to essential resources. This project starts with a manageable piece of that larger challenge: identifying where reported outages may worsen. Comparing quantum and conventional methods helps assess what the quantum approach contributes, while building my understanding of its capabilities and limitations.

Data and feasibility checks

539,068	119,358	47,849	501
South Carolina records retained	explicit zero counts	windows with all-positive reports	escalation windows

ORNL EAGLE-I provides county-level reported customer outage counts at approximately 15-minute intervals. The study downloaded the 2022 annual file from the authors' Figshare repository and retained 539,068 South Carolina records, with zero conflicting duplicate rows identified by the implemented county–timestamp check. Customer counts are not counts of individual people. [1, 2]

The downloaded South Carolina records contained 119,358 explicit zero counts, although the data publication describes zero records as omitted. Missing timestamps were never filled with zeros. The main experiment conservatively required positive counts at every interval in each eligible history-and-outcome window; this avoids depending on the interpretation of explicit zeros but restricts the study population.

An initial audit allowed complete reporting windows containing zeros and found 78,020 eligible hourly windows, including 506 escalation windows at the chosen threshold. Requiring all reports to be positive retained 47,849 windows and 501 escalation windows, grouped descriptively into 350 county episodes

across 153 episode-start dates. Grouping combined flagged hours separated by no more than 24 hours within a county. These are not independently verified outages or independent storms.

Continuous county records do not guarantee complete reporting from every utility. The dataset cannot establish whether a change arose from new physical outages, reporting changes, or corrections.

Workflow

Outcome and inputs

A prediction time was eligible only when positive reports existed at every 15-minute interval from 120 minutes before prediction through 75 minutes afterward.

An escalation required reported customer outages at both 60 and 75 minutes afterward to exceed the prediction-time count by at least 500 customers and at least 50%. The increase can begin earlier than 60 minutes, so this is not a guaranteed one-hour warning before onset.

Four outage inputs were the natural logarithm of one plus the customer count at prediction time and 30, 60, and 120 minutes beforehand.

Two weather inputs indicated whether an original severe-thunderstorm warning or tornado warning naming the county had been issued 15–120 minutes before prediction. The 15-minute delay is an assumed availability buffer, not a measured archive-delivery latency. Other hazards and warnings outside that interval were excluded.

Warning extraction and timing

The archive's CSV uses fields including UTC_PRODISSUE, UGC, and PRODUCT_ID. Because some archived event metadata reflects later updates, the analysis retrieved original warning texts and extracted county codes from messages carrying NEW severe-thunderstorm or tornado warning codes. Message identifiers supplied their original issuance times. Updated expiration times, later damage information, and later warning-area changes were not predictors. [3]

For January–September, extraction produced 1,269 county-warning records. All 60 unmatched messages were verified to name only counties outside South Carolina; some named both Georgia and North Carolina. The October–December query returned five storm-warning messages, all parsed without unresolved cases.

Chronological partitions

Partition	Period	Eligible windows	Escalation windows
Training	January–June, excluding June 30	22,445	213
Validation	July–September, excluding September 30	12,323	135
Final test	October–December	12,438	100

One-day gaps separate partitions to avoid overlapping history/outcome windows. Splits are chronological rather than random. The same storm can still affect adjacent periods or multiple counties.

Validation informed exploratory development. Model configurations were frozen before final-test scoring; no model was retrained on final-test outcomes.

Models and quantum implementation

Logistic regression and random forests used the full training partition, with versions using outage history alone and versions adding warning indicators.

Both SVMs used the same 639 training windows: all 213 escalation windows and 426 randomly sampled non-escalation windows, with seed 42. This training sample intentionally changes the class proportion. Validation and test retained their natural proportions.

Three-fold stratified group cross-validation kept calendar weeks together within the matched training sample. Min–max scaling was fitted separately inside each fold. Average precision selected settings.

The conventional SVM used an RBF kernel. The quantum SVM used Qiskit's zz_feature_map with six qubits, one repetition, and linear entanglement, and FidelityStatevectorKernel for exact simulated state overlaps. The SVM solver remained classical. No physical quantum hardware or hardware-noise experiment was used.

Selected conventional settings were C = 10 and gamma = 0.1. Selected quantum settings were C = 10 and bandwidth = 0.1. Both used balanced class weights.

Evaluation

Average precision evaluates ranking across thresholds; it is not classification accuracy. Final-test escalation prevalence was 0.804%, approximately the uninformative reference for average precision.

A second evaluation selected the highest-ranked 1% of county-hours: 124 validation windows and 125 final-test windows. Equal scores were resolved with a fixed random seed independently of outcome labels. This is retrospective batch prioritization, not a deployable threshold or a real-time staffing policy. Detection counts refer to county-hours, not independent emergencies.

Results

Final chronological test

Method	Training windows	Average precision	Escalation windows detected	False alerts
Recent-increase rule	No fitted model	0.029	9	116
Logistic regression	22,445	0.034	8	117
Random forest	22,445	0.061	15	110
Logistic regression plus warnings	22,445	0.031	4	121
Random forest plus warnings	22,445	0.058	14	111
Matched conventional SVM	639	0.046	11	114
Matched quantum-kernel SVM	639	0.049	13	112

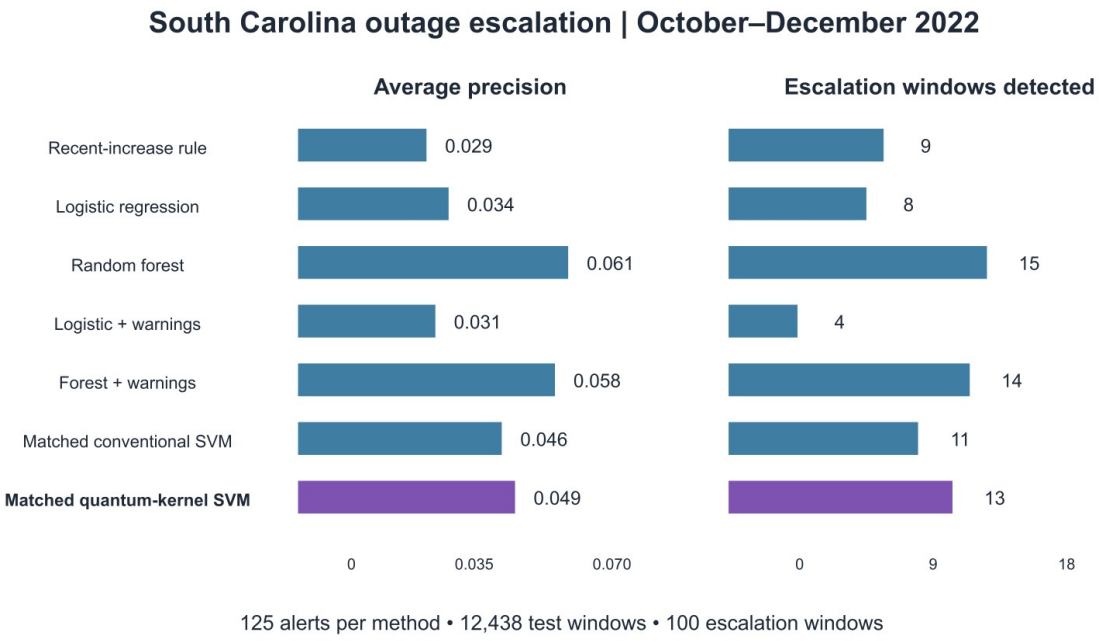
Each method in this table selected 125 of 12,438 test windows. There were 100 escalation windows in the test set, so the detection counts also equal recall percentages.

The quantum model detected two more escalation windows than the matched conventional SVM, with an approximately 0.003 higher average precision at the reported rounding. This is a small observed difference without an uncertainty estimate or a significance test. It does not establish a reliable advantage.

The full-training random forest ranked best overall. Its larger training sample means its comparison with the quantum model combines differences in method and training-data availability. This experiment does not establish sample efficiency; that would require learning curves across multiple training sizes and repeated samples.

Final test comparison figure

The figure compares ranking performance and detections under the common alert budget. The binary warning rule is omitted from this figure because its forced-budget result includes randomly selected unwarned windows.



Weather context and its limits

Period	Recently warned windows	Escalations among warned windows	Rate among warned windows	Rate among unwarned windows
Training	349	22	6.30%	0.86%
Validation	158	14	8.86%	1.00%
Final test	2	0	0%	0.80%

Recent warnings enriched escalation risk in the training and validation samples. The selected warning context was almost absent in the final period, and its association did not translate into improved final-test

results for the augmented conventional models. This finding is specific to the warning types, timing window, and eligible sample; it does not establish that weather is irrelevant.

The warning-only rule generated two actual warning-based alerts and zero detections in the final test. An earlier table forced this binary rule to fill 125 slots and counted one escalation among randomly selected unwarned windows. That detection is not credited to weather warnings, and the forced-budget warning-rule result is excluded from the principal comparison.

Limitations and implications

Operational usefulness remains unproven. The quantum model missed 87 of 100 escalation windows and produced 112 false alerts in the fixed-budget evaluation. No action-time, coordination, cost, or human-outcome benefits were measured.

Reporting selection matters. Eligibility uses future reporting completeness and positive outage counts, selecting periods of ongoing reported outages. Deployment cannot know future reporting completeness. Apparent escalations may include changes in reporting.

Generalization is limited. This is one state and one year. County-hours can be correlated within storms. A chronological split and week-grouped training checks reduce some leakage risks but do not make storm observations independent.

Weather context is narrow. Only recent tornado and severe-thunderstorm warnings were included. County-level warning membership is coarse; it does not imply all residents or infrastructure in a county were exposed. Other hazards, warning lead times, infrastructure conditions, and local support capacity were not modeled.

The quantum comparison is small and simulated. Only one circuit family and a limited parameter grid were studied. No repeated-sample uncertainty analysis, computational speed comparison, hardware experiment, or practical quantum advantage was established.

> **NEXT STEPS**
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> A stronger next investigation would use more years and regions, validation grouped by weather event, broader time-appropriate weather inputs, and uncertainty estimates. Before operational testing, it would require calibrated alert thresholds and a decision-cost framework developed with emergency coordinators and communities. Human benefits would need direct evaluation rather than inference from outage counts.

Conclusion

This initial feasibility study implemented and reproduced a Qiskit quantum-kernel workflow for an energy-related screening task. The quantum SVM showed a modest observed improvement over a conventional SVM trained on identical examples, while a conventional random forest performed better overall.

The results support further investigation of data quality, available warning context, and model generalization. They do not establish that quantum computation is necessary, that quantum advantage was achieved, or that emergency coordination or human outcomes improved.

Reproducibility and acknowledgments

The companion submission notebook begins with a package-installation command and prints package versions. It retrieves the ORNL annual file, filters South Carolina, reconstructs eligibility and labels, obtains original warning messages, trains the comparisons, evaluates the reserved period, and generates labeled figures. Raw-message caching reduces repeated archive requests.

The cleaned notebook was rerun in Colab and reproduced the reported final-test figures on October 24, 2026. External archive contents and unpinned dependency installations may change in future runs.

The QML workflow was adapted from the hackathon's QML_1_Lecture.ipynb. AI assistance supported code development, debugging, documentation, and interpretation. Data collection is credited to ORNL and the NWS warning archive maintained by Iowa State University.

References

1. Tansakul and colleagues. EAGLE-I Power Outage Data 2014–2022. ORNL dataset. Official dataset record (https://doi.org/10.13139/ORNLNCCS/1975202). Figshare distribution used in the notebook (https://doi.org/10.6084/m9.figshare.24237376).
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3. Iowa Environmental Mesonet. NWS VTEC archive documentation (https://mesonet.agron.iastate.edu/info/datasets/vtec.html) and warning download service (https://mesonet.agron.iastate.edu/cgi-bin/request/gis/watchwarn.py?help).
4. IQ Hackathon Fall 2026 Beginners. Training materials and QML notebook (https://github.com/iq-hackathon/IQ-Hackathon-Fall-2026-Beginners).